

WTWIS8 2016 EXAM MEMO

Q1: $\cos^{-1}(2x-1)$

$$-1 \leq 2x-1 \leq 1$$

$$0 \leq 2x \leq 2$$

$$0 \leq x \leq 1$$

$$\therefore [0, 1]$$

$$\therefore c$$

Q2: $\cosh(\ln(3))$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\frac{e^{\ln(3)} + e^{-\ln(3)}}{2}$$

$$= \frac{3 + 1/3}{2}$$

$$= 5/3$$

$$\therefore c$$

Q3: $\left[\frac{1}{2}(1)(2) + (1)(2) + \frac{1}{2}(1)(2) \right] - \frac{1}{2}(1)(1) - (1)(1)$

$$1 + 2 + 1 - \frac{1}{2} - 1$$

$$= \frac{5}{2} = 2.5$$

$$\therefore b$$

Q4: i false

ii true

iii true

$$\therefore d$$

Q5: $f(x) = \sin^{-1}(x^2+x) + s^x$

$$f'(x) = \frac{2x+1}{\sqrt{1-(x^2+x)^2}} + s^x \ln(s)$$

$$f'(0) = 1 + \ln(s)$$

$$\therefore d$$

Q6 $\lim_{t \rightarrow 1^+} \left(\frac{3}{t-1} - \frac{5}{t^2-1} \right) \quad (\infty - \infty)$

$$\lim_{t \rightarrow 1^+} \frac{3}{t-1} - \frac{5}{(t+1)(t-1)}$$

$$\lim_{t \rightarrow 1^+} \frac{3(t+1) - 5}{(t+1)(t-1)}$$

$$\lim_{t \rightarrow 1^+} \frac{3t - 2}{(t+1)(t-1)} \quad \left(\frac{\infty}{\infty} = \pm \infty \right)$$

$$= \infty$$

Q7: $f(x) = x^2 - 4$

VA: $x^2 - 4 \neq 0$

$$(x+2)(x-2) \neq 0$$

$$x \neq 2$$

$$x \neq -2$$

$$\lim_{x \rightarrow 2^+} f(x) = +\infty$$

$$\lim_{x \rightarrow -2^+} f(x) = -\infty$$

$$\lim_{x \rightarrow 2^-} f(x) = -\infty$$

$$\lim_{x \rightarrow -2^-} f(x) = \infty$$

$\therefore$ VA @ $x = 2, x = -2$

b.

Q8: $f'(x) > 0$

$$\frac{-12x}{(x^2-4)^2} > 0$$

$$f'(x) = 0 \quad f' \text{ DNE}$$

$$x = 0$$

$$x = -2, 2$$

-12x	+	-2	+	0	-	2	-
$(x^2-4)^2$	+		+		+		+
	+		+		-		-
	⊕		⊕		⊖		⊖

$$x \in (-\infty, -2) \cup (-2, 0)$$

b.

Q9. $f'' < 0$

$$\frac{12(3x^2+4)}{(x^2-4)^3} < 0$$

$f'' = 0$

f'' DNE

$12(3x^2+4) \neq 0$

$x = 2, x = -2$

$3x^2+4 \neq 0$

$12(3x^2+4)$	$+ \left \begin{array}{c} -2 \\ + \end{array} \right $	$+ \left \begin{array}{c} 2 \\ + \end{array} \right $
$(x^2-4)^3$	$+ \left \begin{array}{c} - \\ + \end{array} \right $	$+ \left \begin{array}{c} - \\ + \end{array} \right $
$---$	$\oplus$	$\ominus$

$x \in (-2, 2)$

$\therefore d$

Q10. $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos(\theta)$

$= (3)(\sqrt{6}) \cos(\pi/4)$

$= (3)(\sqrt{2} \cdot \sqrt{3}) \left(\frac{1}{\sqrt{2}} \right)$

$= (3)(\sqrt{2} \cdot \sqrt{3}) \left(\frac{1}{\sqrt{2}} \right)$

$= 3\sqrt{3}$

$\therefore c$

Section B:

4

$$1.1) \ln(x-2) < 0$$

$$e^{\ln(x-2)} < e^0$$

$$x-2 < 1$$

$$\therefore x < 3$$

and

$$x-2 > 0$$

$$\therefore x > 2$$

$$x \in (2, 3)$$

$$1.2) f(x) = x + \frac{k}{x}$$

$$f'(x) = 1 - \frac{k}{x^2}$$

$$f'(-2) = 0 = 1 - \frac{k}{(-2)^2}$$

$$-1 = -\frac{k}{4}$$

$$\therefore k = 4$$

$$1.3) f'g \cdot g'$$

$$f'(g(1)) \cdot g'(1)$$

$$f'(2) \cdot g'(1)$$

$$5(3)$$

$$= 15$$

$$1.4) \lim_{x \rightarrow 0^-} \frac{x-5}{x} = \infty$$

$$1.5) x^2 + y^2 + z^2 - 2x - 4y + 8z = 15$$

$$x^2 - 2x + y^2 - 4y + z^2 + 8z = 15$$

$$x^2 - 2x + 1 + y^2 - 4y + 4 + z^2 + 8z + 16 = 15 + 1 + 4 + 16$$

$$(x-1)^2 + (y-2)^2 + (z+4)^2 = 36$$

$$\text{centre: } (1, 2, -4)$$

$$r = \sqrt{36} \\ = 6$$

$$1.6) \int (x+1)\sqrt{2x+x^2} dx$$

$$\frac{1}{2} \cdot \frac{2}{3} (2x+x^2)^{3/2} + C$$

$$2 + 2x$$

$$\frac{1}{3} (2x+x^2)^{3/2} + C$$

$$1.7) \int \frac{2x}{1+x^4} dx$$

$$\int \frac{2x}{1+(x^2)^2} dx$$

$$\arctan(x^2) + C$$

$$2.1) y = x^2 \sin(1/x) \quad f'g + fg'$$

$$dy/dx = (2x) \sin(1/x) + x^2 \cos(1/x) (-1/x^2)$$

$$= 2x \sin(1/x) - \cos(1/x)$$

$$2.2) y = \frac{2^{\tan(x)}}{\sec(6x)} \quad \frac{f'g - fg'}{g^2}$$

$$dy/dx = \frac{(2^{\tan(x)} \cdot \sec^2(x) \cdot \ln(2))(\sec(6x)) - 2^{\tan(x)}(\sec(6x) \tan(6x) \cdot 6)}{(\sec(6x))^2}$$

$$2.3) e^{sxy} \cdot y = \sin(y)$$

$$\text{let } y = f(x)$$

$$e^{sx \cdot f(x)} \cdot f(x) = \sin(f(x))$$

$$e^{sx \cdot f(x)} (f(x) + x \cdot f'(x)) + f'(x) = \cos(f(x)) \cdot f'(x)$$

$$e^{sx \cdot f(x)} f(x) + e^{sx \cdot f(x)} x \cdot f'(x) + f'(x) = \cos(f(x)) \cdot f'(x)$$

$$e^{sx \cdot f(x)} f(x) = \cos(f(x)) \cdot f'(x) - e^{sx \cdot f(x)} x \cdot f'(x) - f'(x)$$

$$e^{sx \cdot f(x)} f(x) = f'(x) \left[\cos(f(x)) - e^{sx \cdot f(x)} x - 1 \right]$$

$$\therefore f'(x) = \frac{e^{sx \cdot f(x)} f(x)}{\cos(f(x)) - e^{sx \cdot f(x)} x - 1}$$

$$\frac{dy}{dx} = \frac{e^{sxy} \cdot y}{\cos(y) - x e^{sxy} - 1}$$

$$2.4) y = \frac{e^{2x}}{(x^2+1)^3 (1+\sin(x))^5}$$

$$\ln(y) = \ln\left(\frac{e^{2x}}{(x^2+1)^3 (1+\sin(x))^5}\right)$$

$$\ln(y) = \ln(e^{2x}) - \ln((x^2+1)^3 (1+\sin(x))^5)$$

$$\ln(y) = 2x \ln(e) - [\ln(x^2+1)^3 + \ln(1+\sin(x))^5]$$

$$\ln(y) = 2x - 3 \ln(x^2+1) - 5 \ln(1+\sin(x))$$

$$\frac{1}{y} \frac{dy}{dx} = 2 - 3 \left(\frac{2x}{x^2+1} \right) - 5 \left(\frac{\cos(x)}{1+\sin(x)} \right)$$

$$\frac{dy}{dx} = \frac{e^{2x}}{(x^2+1)^3 (1+\sin(x))^5} \left[2 - \frac{6x}{x^2+1} - \frac{5 \cos(x)}{1+\sin(x)} \right]$$

$$3.1) \int \frac{x^2+1}{2x} dx$$

$$\int \frac{x^2}{2x} dx + \int \frac{1}{2x} dx$$

$$\frac{1}{2} \int x dx + \frac{1}{2} \int \frac{1}{x} dx$$

$$\frac{1}{2} \left[\frac{1}{2} x^2 \right] + \frac{1}{2} \ln|x| + C$$

$$\therefore \frac{1}{4} x^2 + \frac{1}{2} \ln|x| + C$$

$$3.2) f'(t) = \frac{3}{\sqrt{t}}$$

$$f(t) = \int f'(t) = 3 \int t^{-1/2} dt$$

$$= 3(2\sqrt{t}) + C$$

$$f(1) = 9 = 3(2\sqrt{1}) + C$$

$$9 = 6 + C$$

$$C = 3$$

$$\therefore f(t) = 6\sqrt{t} + 3$$

$$3.3) \int \frac{2+x}{\sqrt{1-x^2}} dx$$

$$\int \frac{2}{\sqrt{1-x^2}} dx + \int \frac{x}{\sqrt{1-x^2}} dx$$

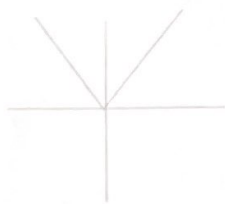
$x(1-x^2)^{-1/2}$

$$2 \arcsin(x) - \left(\frac{1}{2}\right) 2(1-x^2)^{1/2} + C$$

$$-2x$$

$$2 \arcsin(x) - \sqrt{1-x^2} + C$$

$$3.4 \int_{-4}^2 |x| dx$$



$$\int_{-4}^0 -x dx + \int_0^2 x dx$$

$$-\frac{1}{2}x^2 \Big|_{-4}^0 + \frac{1}{2}x^2 \Big|_0^2$$

$$\left[-\frac{1}{2}(0)^2 + \frac{1}{2}(4)^2 \right] + \left[\frac{1}{2}(2)^2 - \frac{1}{2}(0)^2 \right]$$

$$= 10$$

$$\textcircled{4} \lim_{x \rightarrow 0} (1+2x)^{\operatorname{cosec}(x)} \quad (1^\infty)$$

$$+ \operatorname{cosec}(x) = 1/\sin(x)$$

$$\text{let } y = (1+2x)^{\operatorname{cosec}(x)}$$

$$\ln(y) = \operatorname{cosec}(x) \ln(1+2x)$$

$$\lim_{x \rightarrow 0} \ln(y) = \lim_{x \rightarrow 0} \operatorname{cosec}(x) \ln(1+2x) \quad (\infty \times 0)$$

$$\lim_{x \rightarrow 0} \ln(y) = \lim_{x \rightarrow 0} \frac{\ln(1+2x)}{\sin(x)} \quad (0/0)$$

$$\lim_{x \rightarrow 0} \ln(y) \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{\left(\frac{2}{1+2x} \right)}{\cos(x)}$$

$$\lim_{x \rightarrow 0} \ln(y) = 2$$

$$\lim_{x \rightarrow 0} y = e^2$$

$$\therefore \lim_{x \rightarrow 0} (1+2x)^{\operatorname{cosec}(x)} = e^2$$

Q5) $\vec{a} = \langle 3, 2, 1 \rangle$
 $\vec{b} = \langle -1, 1, 0 \rangle$

* if you cross the vectors ($\vec{b} \times \vec{a}$),
 you'll get the same answer just
 $\times \ominus$ (opposite direction)

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 3 & 2 & 1 \\ -1 & 1 & 0 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} i - \begin{vmatrix} 3 & 1 \\ -1 & 0 \end{vmatrix} j + \begin{vmatrix} 3 & 2 \\ -1 & 1 \end{vmatrix} k$$

$$\vec{a} \times \vec{b} = \langle 0-1, -(0-(-1)), 3-(-2) \rangle$$

$$\vec{a} \times \vec{b} = \langle -1, -1, 5 \rangle$$

$$\text{unit vector: } \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{\langle -1, -1, 5 \rangle}{\sqrt{(-1)^2 + (-1)^2 + (5)^2}}$$

$$= \left\langle \frac{-1}{\sqrt{27}}, \frac{-1}{\sqrt{27}}, \frac{5}{\sqrt{27}} \right\rangle$$

G1) $f(x_1) < f(x_2)$, whenever $x_1 < x_2$ in I

G2) let x_1 and x_2 be any two numbers in the interval I , with
 $x_1 < x_2$

we are given $f'(x) > 0$

$\Rightarrow f$ diff on (x_1, x_2) and
 f continuous on $[x_1, x_2]$

By the MVT there exists a $c \in (x_1, x_2)$ so that

$$f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$\Rightarrow f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

Now $f'(c) > 0$ given on $x_2 - x_1 > 0$

$$f(x_2) - f(x_1) > 0$$

$$f(x_2) > f(x_1)$$

$\Rightarrow f$ increasing function on I , since x_1 and x_2 any two numbers.

Q7.1) $f(x) = 1 - e^{-x}$

$y = 1 - e^{-x}$

* to find $f^{-1}(x)$, swap y and x , and solve for y .

$x = 1 - e^{-y}$

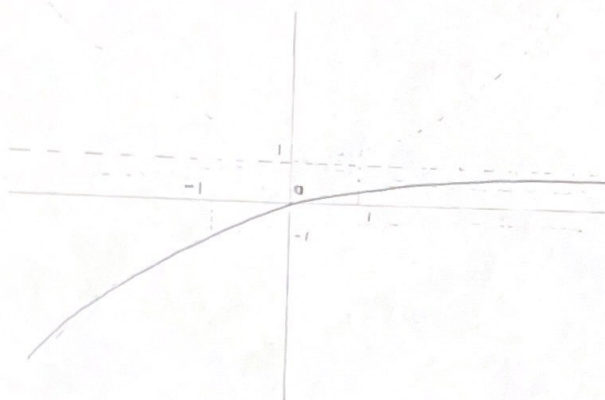
$x - 1 = -e^{-y}$

$-x + 1 = e^{-y}$

$-y = \ln|-x + 1|$

$y = -\ln|1 - x|$

7.2)



7.3) $\int_{-1}^0 1 - e^{-x} dx + \int_0^1 1 - e^{-x} dx$

* area is always +!

7.4. $f(x) = 1 - e^{-x}$ @ point $(a, f(a))$

$f'(x) = e^{-x}$

$f'(a) = e^{-a}$

$y - f(a) = e^{-a}(x - a)$

$y - (1 - e^{-a}) = e^{-a}(x - a)$

passing through $(1, 1)$

$1 - (1 - e^{-a}) = e^{-a}(1 - a)$

$1 - 1 + e^{-a} = e^{-a} - ae^{-a}$

$-ae^{-a} = 0$

$\therefore a = 0$

$y - (1 - 1) = 1(x - 0)$

$y = x$